

1)

The three fundamental theorems for gradient, divergence and curl are:

$$\text{Gradient: } \int_a^b (\nabla f) \cdot dl = f(b) - f(a)$$

$$\text{Divergence: } \int_V (\nabla \cdot A) d\tau = \oint_S A \cdot da$$

$$\text{Curl: } \int_S (\nabla \times A) \cdot da = \oint_L A \cdot dl$$

- Gradient theorem gives that with a function f , the integral of the gradient of f between two points in any given path is the same as the difference of f given the two points. This gives that the potential of a function between two points can be found with any path.
- Divergence theorem gives that a vector A 's flux integral across a surface is the same as the density of flux (divergence) within the surface. This can be seen as the rate of water flowing out of an area's surface is the same as the amount of flow inside the area.
- Curl theorem gives that the flux of the curl through a surface is the same as the line integral across the boundary of the surface. This can be seen as how much swirl a force has in an area is equal to walking around the edge of the area and seeing how much of the force is in line with the edge.

2)

2)

$$\nabla \times \mathbf{v} = \begin{pmatrix} \frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \\ \frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \\ \frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \end{pmatrix} = \begin{pmatrix} -2y \\ -3z \\ -x \end{pmatrix} \quad da = dx dz \hat{y}$$

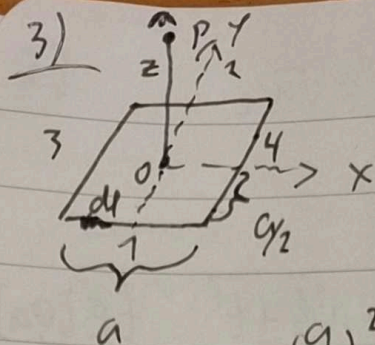
$$\int (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = - \int_0^2 \int_0^{2-z} 3z \, dx \, dz = -3 \int_0^2 z(2-z) \, dz$$

$$= -3 \left[z^2 - \frac{1}{3} z^3 \right]_0^2 = -12 + 8 = \underline{\underline{-4}}$$

$$\oint \mathbf{v} \cdot d\mathbf{l} = \int_0^1 2t \cdot 0 \cdot 2 \, dt + \int_0^1 2 \cdot 0 \cdot (2-2t) \cdot 2 \, dt \Big|_{(x,y,z) \Rightarrow (2t, 0, 2-2t)} \\ - 3 \int_0^1 (2-2t) \cdot 2t \cdot 2 \, dt = -24 \int_0^1 t - t^2 \, dt = -24 \left(\frac{1}{2} - \frac{1}{3} \right) = \underline{\underline{-4}} \quad t \in [0, 1]$$

(Convert from (x, y, z) to t based to calculate diagonal line, vertical and horizontal lines, cancel since $y=0$.)

3)



$$\mathbf{r} = -x\hat{x} + \left(\frac{a}{2}\right)\hat{y} + z\hat{z}$$

$$|\mathbf{r}| = \sqrt{x^2 + \left(\frac{a}{2}\right)^2 + z^2}$$

$$E = \frac{1}{4\pi\epsilon_0} \int \frac{d\mathbf{r}}{|\mathbf{r}|^2} dl$$

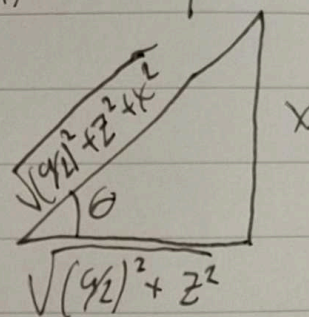
$$E_1 = \frac{1}{4\pi\epsilon_0} \int_{-a/2}^{a/2} \frac{\lambda(-x\hat{x} + \frac{a}{2}\hat{y} + z\hat{z})}{(x^2 + (\frac{a}{2})^2 + z^2)^{3/2}} dx$$

$$E_1 = \frac{\lambda}{4\pi\epsilon_0} \left[-\hat{x} \int_{-a/2}^{a/2} \frac{x dx}{(x^2 + (\frac{a}{2})^2 + z^2)^{3/2}} + \left(\frac{a}{2}\hat{y} + z\hat{z}\right) \int_{-a/2}^{a/2} \frac{dx}{(x^2 + (\frac{a}{2})^2 + z^2)^{3/2}} \right]$$

$$\Rightarrow \frac{\lambda}{4\pi\epsilon_0} \left[0 + \frac{(\frac{a}{2}\hat{y} + z\hat{z})}{((\frac{a}{2})^2 + z^2)^{3/2}} \int_{-a/2}^{a/2} \frac{dx}{\left(\frac{x^2}{(\frac{a}{2})^2 + z^2} + 1\right)^{3/2}} \right] \quad \left| \begin{array}{l} \tan\theta = \frac{x}{\sqrt{(\frac{a}{2})^2 + z^2}} \\ dx = \sqrt{(\frac{a}{2})^2 + z^2} \sec^2\theta \end{array} \right.$$

$$\Rightarrow \frac{\lambda}{4\pi\epsilon_0} \frac{(\frac{a}{2}\hat{y} + z\hat{z})}{((\frac{a}{2})^2 + z^2)} \cdot \int_{-a/2}^{a/2} \cos\theta d\theta$$

$$\sin\theta = \frac{x}{\sqrt{(\frac{a}{2})^2 + z^2 + x^2}} \quad \left| \begin{array}{l} \text{at } x = a/2, \theta = \theta_1 \\ \text{at } x = -a/2, \theta = \theta_2 \end{array} \right.$$

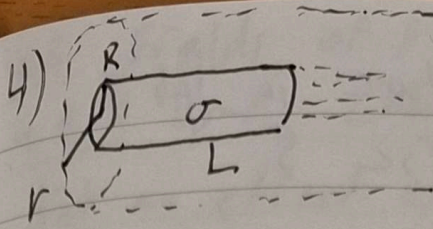


$$\Rightarrow E_1 = \frac{\lambda}{4\pi\epsilon_0} \frac{(\frac{a}{2}\hat{y} + z\hat{z})a}{((\frac{a}{2})^2 + z^2)\sqrt{\frac{a}{2} + z^2}}$$

E_2 's $\hat{y}$ component will cancel E_1 's and E_3 & E_4 will do same for $\hat{x}$.

$$\Rightarrow E_{\text{tot}} = 4 \cdot E_{1z} = \frac{\lambda}{\pi\epsilon_0} \frac{za\hat{z}}{((\frac{a}{2})^2 + z^2)\sqrt{\frac{a}{2} + z^2}}$$

4)



Gauss Law

$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{\text{enc}}}{\epsilon_0} \quad r < R \Rightarrow 0$$

$r > R$: Flat surface cancel, only look radially out
no charge enclosed

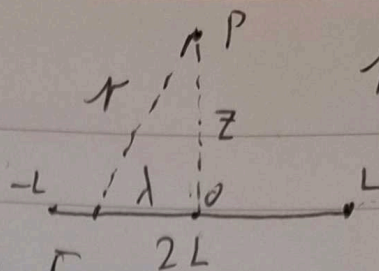
$$\Rightarrow \int \mathbf{E} \cdot d\mathbf{a} = \int |E| da \cos 0 = |E| \int da = |E| 2\pi RL$$

$$\Rightarrow Q_{\text{enc}} = \int \sigma da = 2\pi r L \sigma$$

$$\Rightarrow |E| 2\pi RL = \frac{\sigma}{\epsilon_0} 2\pi r L \Rightarrow \underline{\underline{E = \frac{\sigma r}{\epsilon_0 R}}}$$

5)

5) $V = \frac{1}{4\pi\epsilon_0} \int \frac{\lambda}{r} dl$



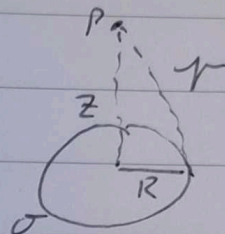
$$r = \sqrt{z^2 + x^2}$$

$$\Rightarrow V = \frac{1}{4\pi\epsilon_0} \int_{-L}^L \frac{\lambda dx}{\sqrt{z^2 + x^2}} = \frac{\lambda}{4\pi\epsilon_0} \left[\ln(x + \sqrt{z^2 + x^2}) \right]_{x=-L}^{x=L}$$

$$\Rightarrow V = \frac{\lambda}{4\pi\epsilon_0} \left(\ln \left(\frac{L + \sqrt{z^2 + L^2}}{\sqrt{z^2 + L^2} - L} \right) \right)$$

$$\underline{\underline{E = -\nabla V = \begin{pmatrix} 0 \\ 0 \\ -\frac{d}{dz} V \end{pmatrix} = \frac{\lambda}{4\pi\epsilon_0} \frac{2zL}{\sqrt{z^2 + L^2} (\sqrt{z^2 + L^2} - L)^2}}}$$

6) $V = \frac{1}{4\pi\epsilon_0} \int \frac{\sigma}{r} d\alpha$



$$r = \sqrt{z^2 + R'^2}$$

$$\Rightarrow V = \frac{1}{4\pi\epsilon_0} \int_0^{2\pi} \int_0^R \frac{\sigma \cdot R'}{\sqrt{z^2 + R'^2}} dR' d\theta$$

$$\Rightarrow V = \frac{\sigma}{2\epsilon_0} \left[\sqrt{z^2 + R^2} \right]$$

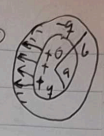
$$\underline{\underline{E = \begin{pmatrix} 0 \\ 0 \\ -\frac{d}{dz} V \end{pmatrix} = \frac{\sigma}{2\epsilon_0} \left(\frac{z}{\sqrt{z^2 + R^2}} \right)}}$$

6) a) i) $W_1 = \frac{1}{2} \sum_{i=1}^3 q_i V(r_i) = \frac{1}{2} \cdot (-q) \left(\frac{1}{4\pi\epsilon_0} \left(\frac{q}{2a} - \frac{q}{2a\sqrt{2}} \right) \right)$
 ii) $W_2 = \frac{1}{2} q \cdot V(2a) = \frac{1}{2} q \left(\frac{1}{4\pi\epsilon_0} \cdot \frac{-q}{2a} \right) = \frac{q^2}{4\pi\epsilon_0 a} \left(-\frac{1}{4} \right)$

$W_1 = 0$ $W_{\text{tot}} = W_1 + W_2 + W_3$
 $\Rightarrow 0 + \frac{q^2}{4\pi\epsilon_0 a} \left(-\frac{1}{4} - \frac{1}{4} + \frac{1}{4\sqrt{2}} \right) = \frac{q^2}{4\pi\epsilon_0 a} \left(\frac{1}{4\sqrt{2}} - \frac{1}{2} \right)$

b) i) $W_3 = \frac{1}{2} \sum_{i=1}^3 q_i V(r_i) = -\frac{q}{2} \left(\frac{1}{4\pi\epsilon_0} \left(\frac{q}{2a} - \frac{q}{2a\sqrt{2}} \right) \right)$

ii) W_1 and W_2 is same as in a)
 $\Rightarrow W_{\text{tot}} = \frac{q^2}{4\pi\epsilon_0 a} \left(-\frac{1}{4} - \frac{1}{2} + \frac{1}{2\sqrt{2}} \right) = \frac{q^2}{4\pi\epsilon_0 a} \left(\frac{1}{2\sqrt{2}} - \frac{3}{4} \right)$

7) a) $W = \frac{\epsilon_0}{2} \int E^2 d\tau \Rightarrow E_a = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$ 
 $\Rightarrow E_b = \frac{1}{4\pi\epsilon_0} \frac{-q}{r^2}$
 E -field outside and inside sphere = 0

$\Rightarrow W_{\text{bf}} = \frac{\epsilon_0}{2} \frac{1}{(4\pi\epsilon_0)^2} \int_0^{2\pi} \int_0^\pi \int_a^\infty \frac{q^2}{r^4} \cdot r^2 \sin\theta dr d\theta d\phi$

$\Rightarrow W_{\text{tot}} = \frac{q^2}{8\pi\epsilon_0} \int_a^\infty \frac{1}{r^2} dr = \frac{q^2}{8\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right)$

b) $W_a = \frac{\epsilon_0}{2} \frac{1}{(4\pi\epsilon_0)^2} \int_0^{2\pi} \int_0^\pi \int_a^\infty \frac{q^2}{r^4} \cdot r^2 \sin\theta dr d\theta d\phi = \frac{q^2}{8\pi\epsilon_0} \cdot \frac{1}{a}$
 $W_b = \frac{q^2}{8\pi\epsilon_0} \cdot \frac{1}{b}$

$W_{a,b} = \frac{\epsilon_0}{2} \int (E_a \cdot E_b) d\tau = \epsilon_0 \int_0^{2\pi} \int_0^\pi \int_a^\infty \frac{q}{4\pi\epsilon_0 r^2} \cdot \frac{-q}{4\pi\epsilon_0 r^2} \cdot r^2 \sin\theta dr d\theta d\phi$
 $\Rightarrow -\frac{q^2}{8\pi\epsilon_0} \cdot \frac{1}{b}$

$\Rightarrow W_a + W_b + W_{a,b} = \frac{q^2}{8\pi\epsilon_0} \left(\frac{1}{a} + \frac{1}{b} - \frac{2}{b} \right) = \frac{q^2}{8\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right)$

8)

$$b) a) \oint E \cdot d\vec{s} = \frac{Q_{enc}}{\epsilon_0} \Rightarrow |E| 2\pi r L = \frac{Q}{\epsilon_0}$$

$$\Rightarrow \underline{\underline{E = \frac{Q}{2\pi\epsilon_0 r L} \hat{r}}}$$

$$b) V = - \int_a^b E \cdot d\vec{l} = \frac{-Q}{2\pi\epsilon_0 L} \int_a^b \frac{1}{r} dr = \underline{\underline{\frac{Q}{2\pi\epsilon_0 L} \ln\left(\frac{b}{a}\right)}}$$

$$c) C = \frac{Q}{V} = \frac{2\pi\epsilon_0 L}{\ln(b/a)} \quad \underline{\underline{C/L = \frac{2\pi\epsilon_0}{\ln(b/a)}}}$$

$$d) \ln(b/a) = \frac{2\pi\epsilon_0}{C/L} \Rightarrow a = b \cdot e^{-\frac{2\pi\epsilon_0}{C/L}} \Rightarrow \underline{\underline{16,12}}$$